The S_3-symmetric 3HDM in the soft-broken vacuum: a decoupling-like spectrum, and why the light scalars of the exact-S_3 scan do not survive

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Abstract

The exact-S_3 potential forces the vacuum onto the $\sqrt{3}$ alignment. That alignment forces $V_{us} = 0$, and soft S_3 breaking is what lifts it (companion fermion report). I have re-run the full scalar parameter scan in the soft-broken vacuum, with the same theory and collider cuts, over a 13-dimensional space: eight quartics, two vacuum angles and three free soft masses. 2021 of 1.1×10^8 sampled points survive. The spectrum changes character. The SM-like state is the lightest CP-even scalar in 99.7% of points, the heavy states are near-degenerate and reach the TeV scale, and the light scalars that 16% of the exact-S_3 points carry drop to 0.7%. Those light states are a property of the measure-zero exact slice, not of the S_3 quartics. Along the way, two structural results fix how the scan has to be set up: a coordinate pole in the natural parametrisation of the tadpoles, and a fundamental domain for the vacuum angle.

1 Summary

Everything below comes from `research/thdm_s3/05_soft_scalar_scan.ipynb`; the scan itself is `scan_soft.py`, and its output is `results/viable_points_soft.json`.

1. **Soft does not mean small** (§2). The soft terms are sampled wide, and how close a point is to exact S_3 is read off afterwards instead of being imposed.
2. **A parametrisation pole, avoided**: solving the tadpoles for the wrong soft term divides by $v_1^2 - v_2^2$ (§3).
3. $\varphi \in (0, \pi/3)$ **is a fundamental domain**, and our quark-fit vacuum sits inside it at $\varphi = 53.5°$ (§4).
4. **The spectrum machinery generalises**: the Goldstones stay isolated, and the CP-even sector becomes a full 3×3 with no gauge-phobic state (§5).
5. **The soft-broken spectrum is decoupling-like**, and the exact scan's light scalars do not survive generic soft breaking (§7). This is the main result.

2 What "soft" means

The soft-broken potential adds the four CP-conserving, gauge-invariant quadratics that break S_3:

$$V_{\text{soft}} = m_{D1}^2(x_{11} - x_{22}) - m_{D2}^2(x_{12} + x_{21}) + m_{S1}^2(x_{S1} + x_{1S}) + m_{S2}^2(x_{S2} + x_{2S}).$$

These are the **2** of $\mathbf{2}\otimes\mathbf{2}$ and the doublet $H_S^\dagger H_i$, each contracted with a constant "spurion" doublet of mass^2 coefficients.

Soft refers to operator dimension, not to size. The quartics (and the Yukawas) stay exactly S_3-invariant, and the spurions carry mass dimension two. No loop can therefore generate an S_3-breaking quartic, and the soft terms renormalise only in proportion to themselves. Setting them to zero restores S_3, so small values are technically natural [1]. Natural means stable, not required: soft terms of order v^2 or $(\text{TeV})^2$ are just as consistent. I therefore sampled the three free soft masses as signed uniform variables up to $(500\ \text{GeV})^2$ and tagged each point with the size of its breaking.

Boundedness and unitarity carry over unchanged, since they depend on the quartics alone. The corrected boundedness condition, the tree-unitarity cut and the 12-dimensional backstop of the scalar-sector report all apply to λ exactly as before.

3 The released tadpoles

With the soft terms, the three tadpole conditions become ordinary equations for three unknowns. Nothing forces the alignment any more, and a general (v_1, v_2, v_S) is a stationary point. Which soft term to solve for is **not** a free choice:

- solving for $(\mu_0^2, \mu_1^2, m_{D2}^2)$ divides by $v_1^2 - v_2^2$, a pole on the $\varphi = 45°$ line in the middle of the domain;
- solving for $(\mu_0^2, \mu_1^2, m_{D1}^2)$ divides only by $v_1 v_2$ (and μ_0^2 by v_S), which is regular throughout.

The reason is geometric. The (m_{D2}^2, μ_1^2) tadpole directions, $\propto (v_2, v_1)$ and $\propto (v_1, v_2)$, become parallel at $v_1 = v_2$. The (m_{D1}^2, μ_1^2) directions, $(v_1, -v_2)$ and (v_1, v_2), are parallel only when $v_1 v_2 = 0$. Nothing physical happens at 45°; it is a coordinate singularity. The scan solves for m_{D1}^2, and so does every notebook; it is also the code's default, since solving for m_{D2}^2 puts this pole inside the domain.

The exact model is a limit. With the free soft terms set to zero and the vacuum on $v_2 = \sqrt{3} v_1$, the solved m_{D1}^2 vanishes identically. A falsification check confirms it is non-zero off the alignment. Notebook 01's model is the $\varphi = 60°$, zero-soft slice of this one.

4 A fundamental domain for the vacuum angle

The vacuum now has two angles: $v_1 = v \sin\theta \cos\varphi$, $v_2 = v \sin\theta \sin\varphi$, $v_S = v \cos\theta$. Not every φ is new physics. An S_3 transformation of the doublet, applied together with the same transformation of the spurions, maps the potential to itself. In our basis the element ab is the reflection about the 60° axis,

$$P = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}, \qquad \varphi \mapsto \tfrac{2\pi}{3} - \varphi.$$

The same P acts on *both* spurion doublets, the **2** pair included, even though that pair is quadratic in the fields. Over 200 random image pairs, the transformed point reproduces all seven masses^2 to 2×10^{-13} and the solved m_{D1}^2 to 10^{-12}. Leaving the **2** pair untransformed breaks the agreement, so the check has teeth. Together with b (the reflection about 0°) this makes $\varphi \in (0, \pi/3)$ a fundamental domain. Both of its edges are residual-$\mathbb{Z}_2$ directions, and $\varphi = 60°$ is the exact-S_3 alignment.

The link to the fermion sector, with one subtlety. The fermion analysis depends on the vacuum through the draft-basis ratio $r = v_1/v_2 = -\cot(\varphi - \pi/6)$, checked against the verified basis map. That analysis works at a fixed fermion assignment, so a vacuum and its S_3 image appear there as two different ratios, $r(\varphi)$ and $-\tan\varphi$, with different Yukawas but the same physics. Each scanned point stands for both, and the two coincide (at $|r| = \sqrt{3}$) only on the exact vacuum. Our quark fit's $|r| = 1.354$ corresponds to $\varphi = 66.6°$, outside the domain. Its image inside the domain is $\varphi = 53.5°$, where $\tan\varphi = 1.354$.

5 The spectrum for a general vacuum

The geometric rotation R [2] is built from the vacuum geometry alone, and its first column is the vacuum direction for any (θ, φ). So $R^{\mathsf{T}} M_A R$ and $R^{\mathsf{T}} M_C R$ still have a vanishing first row: the Goldstones remain isolated. I checked this with *exact* arithmetic at a rational point, and the zeros are exact. Everything else changes:

- the leftover CP-odd and charged 2×2 blocks are no longer diagonal (closed-form eigenvalues);
- the CP-even sector is a full 3×3. **No state is protected from the vacuum direction**, so every CP-even state couples to VV, with coupling^2 equal to its overlap^2 with $\hat{v}$. The three sum to the SM value.

Validation: at 60 random soft-broken points, the masses agree with an independent 30-digit diagonalisation of the un-rotated matrices to 2×10^{-15} in all three sectors. The hVV couplings agree with the eigenvector projections to 3×10^{-15}, and the sum rule holds to 10^{-15}. On the aligned, zero-soft slice the soft spectrum reproduces the exact-S_3 one over 2×10^4 points (to 6×10^{-11}), including the exactly gauge-phobic state.

6 The scan

The prior is a choice, recorded in the output file:

input	prior
$\lambda_{1..8}$	$U(-2, 2)$, as in the exact scan
θ	$U(0.05,\ \pi/2 - 0.05)$, as in the exact scan
φ	$U(0,\ \pi/3)$, the fundamental domain
$m^2_{D2},\ m^2_{S1},\ m^2_{S2}$	$U(-M^2, M^2)$, $M = 500$ GeV, signed
$\mu^2_0,\ \mu^2_1,\ m^2_{D1}$	fixed by the tadpoles

The vacuum is required to be a local minimum (no tachyons). The cuts are the exact scan's, in the same order:

cut	exact S_3 surviving	exact S_3 fraction	soft-broken surviving	soft-broken fraction
sampled	6.0×10^7	1	1.1×10^8	1
+ strict BFB	3 092 767	5.15%	5 666 092	5.15%
+ tree unitarity	3 092 767	5.15%	5 666 092	5.15%
+ no tachyons	231 516	0.386%	1 042 644	0.948%
+ $m_{H^\pm} > 80$ GeV	211 193	0.352%	1 035 278	0.941%
+ SM-like at 125 (coupling)	1 994	3.3×10^{-5}	2 170	2.0×10^{-5}
+ LEP neutral veto	1 987		2 170	
+ 12-d BFB backstop	**1 964**	3.3×10^{-5}	**2 021**	1.8×10^{-5}

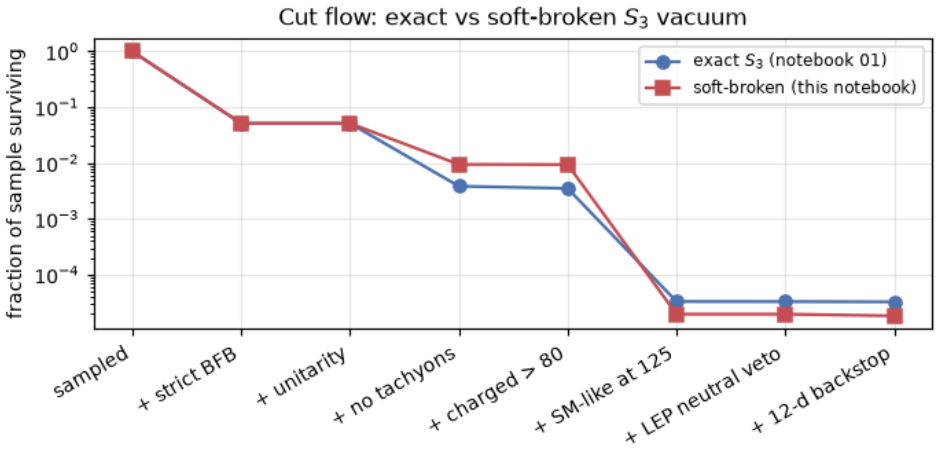


Figure 1: Cut flow, as fractions of each sample: the exact-S_3 scan against the soft-broken one.

The quartic cuts remove the same 95% in both scans, as they must. Two differences matter.

- **No tachyons passes 2.5× more often.** In the exact model every mass^2 is a quartic times v^2, so a λ point with a negative combination is tachyonic for any θ. Soft terms add mass^2 of their own and lift those directions.
- **The coupling cut is where the soft space dies.** The survival fraction per sample is lower in the end, which reflects the extra prior volume as much as physics.

Chunk 0 of the cached scan is reproduced exactly by a fresh run in the notebook. Every chunk-0 survivor's CP-even spectrum is re-derived independently from the symbolic mass matrices.

7 Results

7.1 A decoupling-like spectrum

soft-broken [GeV]				exact S_3 [GeV]			
state	min	median	max	state	min	median	max
h_1	34	125	128	H_2	7	125	128
h_2	124	588	2419	h_0	5	326	521
h_3	382	983	6×10^4	H_1	122	370	849
A_1	55	595	2419	A_1	2	333	671
A_2	304	990	6×10^4	A_2	24	327	708
$H_1^\pm$	85	555	2414	$H_1^\pm$	81	241	470
$H_2^\pm$	246	968	6×10^4	$H_2^\pm$	81	254	743

Soft-broken states are mass-ordered within each sector. Rows are paired by role, not by name.

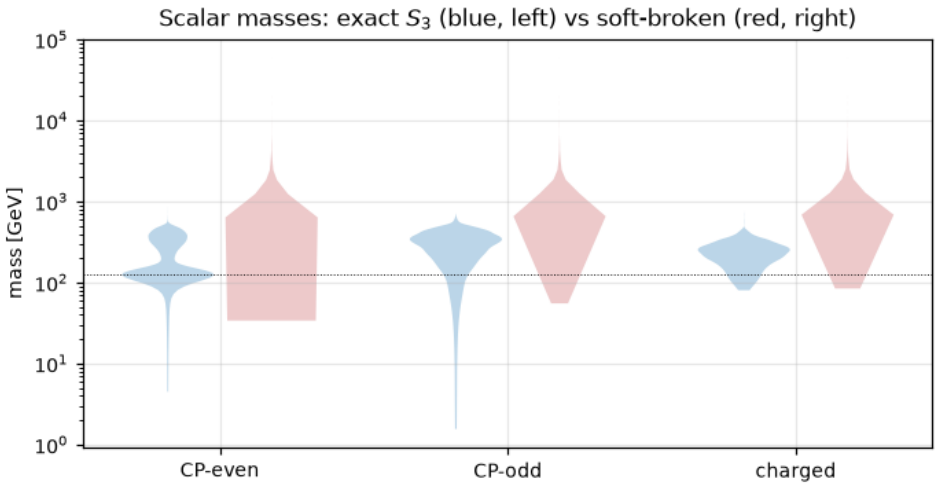


Figure 2: Scalar masses by sector: exact S_3 (blue, left) against soft-broken (red, right). The dotted line is 125 GeV.

In the exact scan every mass^2 is $\lambda \times v^2$ with $|\lambda| \leq 2$, so nothing goes much beyond 850 GeV. With soft terms the heavy states are set by the soft scale. The median heaviest scalar is 1.0 TeV, against 0.44 TeV exact, and half the points have one above 1 TeV. The heaviest CP-even, CP-odd and charged states are near-degenerate: their relative spread has a median of 3.7%, and 1.5% for $h_3 > 1$ TeV. That is the pattern of a heavy doublet whose mass comes from one large soft term, the analogue of the 2HDM's large-m_{12}^2 regime.

7.2 Every state couples to VV, but the non-SM ones barely

- The SM-like state is h_1, the *lightest* CP-even state, in 2015 of 2021 points (and h_2 in the other 6). Its κ^2 has a median of 0.998.
- The other two states couple at the 10^{-4} level (median 4.0×10^{-4}, 90% below 1.5×10^{-2}, maximum 0.099).
- Only 6 points have a CP-even state below the SM-like one, all with $\xi^2 \leq 0.036$, under the LEP floor.

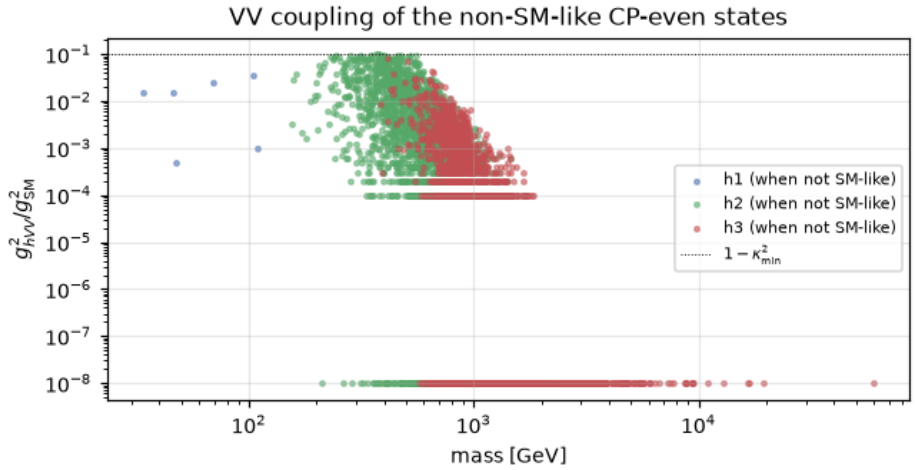


Figure 3: hVV coupling^2 (in SM units) of the non-SM-like CP-even states, against their mass. The dotted line is $1-\kappa^2_{\min}=0.1$, the most the cut allows them to carry. The row at 10^{-8} collects couplings below the stored precision (10^{-4}, the rounding in `viable_points_soft.json`), clipped onto the axis.

7.3 The light scalars of the exact scan do not survive

A non-SM scalar below 100 GeV appears in 15.8% of the exact-S_3 points and in 0.7% of the soft-broken ones. In the exact scan such states are allowed by LEP because h_0 is exactly gauge-phobic and the CP-odd and charged states do not couple to ZZ. With the vacuum released, no state is gauge-phobic, and generic soft breaking lifts them. The light states are a property of the measure-zero exact slice, not of the S_3 quartics. This matters for any phenomenology built on light S_3 scalars.

A side result: $\lambda_4>0$ holds in *every* exact-scan point and in 50.6% of the soft-broken ones. With the alignment and $v_{1,2}>0$ fixed, the $\lambda_4<0$ aligned vacua are all removed by the cuts. This is observed over the 1964 points, not derived.

7.4 How much breaking do viable points need?

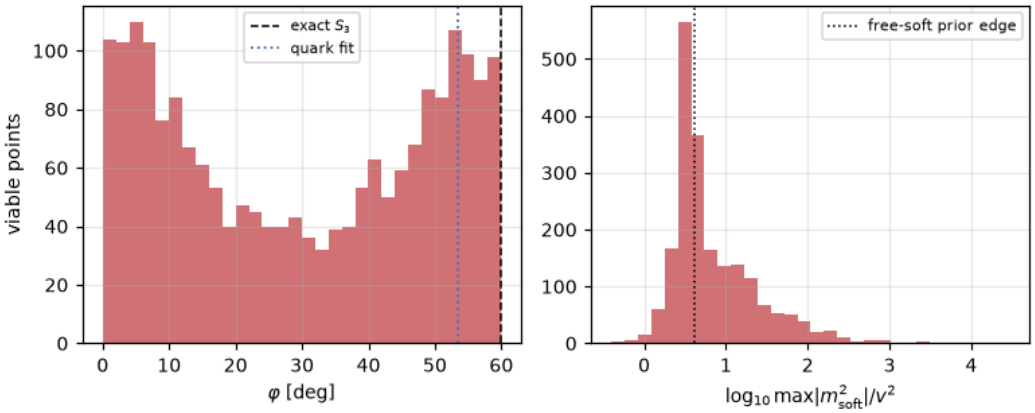


Figure 4: Left: survivors against φ (uniform prior), with the exact-S_3 vacuum (dashed) and the quark-fit vacuum (dotted). Right: the size of the breaking, $\log_{10}\max|m^2_{\rm soft}|/v^2$ over all four soft terms, with the edge of the free-soft prior.

- **No viable point is close to exact S_3 in both senses at once.** The largest soft term is at least $0.40\,v^2$, with a median of $4.1\,v^2$. Only 15 points keep every soft term below v^2. Their spectra run from about 85 GeV to 630 GeV, like the exact scan's heavy states but without its very light ones. Some of this is the prior, since a uniform box in m^2 gives little volume to small values. What the scan does show is that small breaking is *allowed*, including within a few degrees of 60°, and that the viable region does not collapse onto the exact vacuum. It only becomes rare under this measure.
- **Survivors pile up at both ends of the domain** (261 in the first 5°, 237 in the last, about 95 per bin in the middle). The $\varphi \to 0$ end is the decoupling tail: the solved $m^2_{D1} \propto 1/(v_1 v_2)$ blows up there, and all 34 points with a state above 5 TeV have $\varphi < 7°$ (Fig. 5). The $\varphi \to 60°$ end has moderate breaking (median $3.5\,v^2$ within 10°).
- **The quark-fit vacuum is well populated**: 203 points lie within ±2° of $\varphi = 53.5°$.

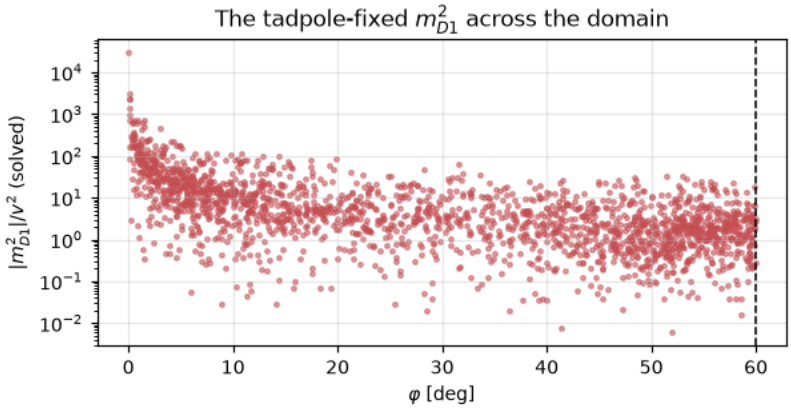


Figure 5: The tadpole-fixed $|m^2_{D1}|/v^2$ across the domain. It grows without bound as $\varphi \to 0$, which is where the multi-TeV tail comes from.

8 Open questions, and what would help

1. **The near-S_3 corner.** A uniform prior in m^2 leaves only 15 points with every soft term below v^2. Whether the light gauge-phobic scalars reappear continuously as the breaking shrinks needs a targeted scan, with log-uniform soft terms and φ near $\pi/3$.

2. **Global minimum.** The scan requires only a local minimum. With soft terms the potential can have several minima, so a multi-start minimisation on the survivors is the missing vacuum-stability cut.

3. **Decays and LFV on the soft points.** The decays analysis parametrises the CP-even sector by one angle δ, which assumes the aligned 2×2 block. The soft points carry a full 3×3 mixing matrix (stored per point), so the couplings and rates have to be regenerated. This is also where the quark channels can finally enter, since the soft-broken quark sector works.

4. **Which vacuum do we want to present?** The quark fit picks a region of the domain that is well populated here. Is the combined statement worth making in the LFV paper: a soft-broken vacuum that reproduces CKM magnitudes, passes the scalar cuts, and has a decoupling-like spectrum? Your call.

A The derivation ledger

The full typeset version is the companion document `derivations_05.tex`; in summary:

claim	status	independent checks
The tadpoles fix $(\mu_0^2, \mu_1^2, m_{D1}^2)$ for a general vacuum	derived	solving for m_{D2}^2 instead is singular at $\varphi = 45°$; $m_{D1}^2 = 0$ on the aligned, zero-soft vacuum
Goldstones isolated; CP-even 3×3 with no gauge-phobic state	derived	exact zeros at a rational point; 30-digit brute force at 60 points; sum rule to 10^{-15}
$\varphi \in (0, \pi/3)$ is a fundamental domain	derived	identical spectra at 200 image pairs; fails if the **2** spurion is left untransformed
$r = -\cot(\varphi - \pi/6)$	derived	agrees with the verified basis map at 40 angles; $\lvert r(\pi/3)\rvert = \sqrt{3}$
Soft terms need not be small	**imported** [1]	technical naturalness

References

[1] G. 't Hooft, *Naturalness, chiral symmetry, and spontaneous chiral symmetry breaking*, NATO Sci. Ser. B **59**, 135 (1980), doi:10.1007/978-1-4684-7571-5_9.

[2] M. Gómez-Bock, M. Mondragón and A. Pérez-Martínez, *Scalar and gauge sectors in the 3-Higgs Doublet Model under the S_3-symmetry*, Eur. Phys. J. C **81**, 942 (2021), arXiv:2102.02800, doi:10.1140/epjc/s10052-021-09731-3.